

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{G}_{xy}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ -G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{Q}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{yzx} \end{bmatrix}$$

$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{yzx} = 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

$$G_{zxy} = -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$

$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$

$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxx} \\ 0 & 0 & 0 & 0 & 0 & G_{yyyy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxx} \\ 0 & 0 & 0 & 0 & G_{yyzz} & 0 \\ 0 & 0 & 0 & G_{yyzz} & 0 & 0 \\ G_{xxxx} & G_{yyyy} & G_{zzxx} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} + 2\mathbb{G}_z^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & 0 & 0 \\ -G_{xxxy} & -G_{yyxy} & -G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_z^{(1)} - 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$

$$G_{yyxy} = \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_z^{\beta(1)}$$

$$G_{yzzx} = -\mathbb{Q}_z^{(1)} - 4\mathbb{Q}_z^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ -G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = -\mathbb{Q}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxxy} \\ 0 & 0 & G_{yyxy} \\ 0 & 0 & G_{zzxy} \\ 0 & G_{yzzx} & 0 \\ G_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = -2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{yyxy} = 2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{zzxy} = \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{(4)} + 2\mathbb{Q}_z^{\alpha(2)}$$

$$G_{yzzx} = -\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{zxyz} = \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & 0 \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{yzzx} &= \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{zxyz} &= -\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzx} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & G_{xxxy} & 0 & 0 & G_{xxyx} \\
0 & 0 & 0 & 0 & 0 & G_{yyxy} & 0 & 0 & G_{yyyx} \\
0 & 0 & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & G_{zzyx} \\
0 & 0 & 0 & 0 & G_{yzxz} & 0 & 0 & G_{yzxz} & 0 \\
0 & 0 & 0 & G_{zxyz} & 0 & 0 & G_{zxzy} & 0 & 0 \\
G_{xyxx} & G_{xyyy} & G_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxy} &= \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(3)} - 2\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} + \mathbb{Q}_z^{\beta(2)} \\
G_{xxyx} &= \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(3)} - 2\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} - \mathbb{Q}_z^{\beta(2)} \\
G_{yyxy} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} - 2\mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} - \mathbb{Q}_z^{\beta(2)} \\
G_{yyyx} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} - 2\mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} + 2\mathbb{Q}_z^{\beta(1)} + \mathbb{Q}_z^{\beta(2)} \\
G_{zzxy} &= \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{(4)} + 2\mathbb{Q}_z^{\alpha(2)} - 4\mathbb{Q}_z^{\beta(1)} \\
G_{zzyx} &= \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{Q}_z^{(3)} - 2\mathbb{Q}_z^{(4)} - 2\mathbb{Q}_z^{\alpha(2)} - 4\mathbb{Q}_z^{\beta(1)} \\
G_{yzxz} &= \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(4)} - 4\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)} \\
G_{yzxz} &= \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(4)} - 4\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)} \\
G_{zxyz} &= \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_z^{(1)} + \mathbb{Q}_z^{(4)} + 4\mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)} \\
G_{zxzy} &= \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{(4)} + 4\mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)} \\
G_{xyxx} &= \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)} - 2\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} - 2\mathbb{Q}_z^{\beta(1)} \\
G_{xyyy} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_z^{(1)} - 2\mathbb{Q}_z^{\alpha(1)} - 2\mathbb{Q}_z^{\beta(1)} \\
G_{xyzx} &= \mathbb{G}_{xy}^{(1)} - \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} + 4\mathbb{Q}_z^{\beta(1)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & G_{yzxz} & 0 & 0 & G_{yzxz} & 0 \\
0 & 0 & 0 & G_{zxyz} & 0 & 0 & G_{zxzy} & 0 & 0 \\
G_{xyxx} & G_{xyyy} & G_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{yzxz} &= 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzxz} &= -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{zxyz} &= 2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxzy} &= -2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzx} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & G_{xxxy} & 0 \\ G_{yyyyx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzxy} & 0 \\ G_{xxyx} & 0 & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{xyyy} & 0 \\ 0 & 0 & \frac{G_{xxxx}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzxy} \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{yyyyx} = -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{xxyx} = \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{xyyy} = -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)}$$